

**Parental identity and psychological need satisfaction as resources against
parental burnout**

Abstract

High levels of stress experienced in the parental role can lead to parental burnout, a condition with severe consequences for both parents and children. Researchers have begun to document the antecedents of parental burnout, focusing primarily on risk factors. However, little is known about resources that prevent parents from developing parental burnout. Based on identity literature and self-determination theory, the purpose of this study was to examine both the unique and interacting roles of two psychological resources potentially protecting parents against parental burnout, that is, parental identity and psychological need satisfaction. A total of 450 parents (63.8% mothers, M age = 36.53) completed online questionnaires on parental identity, psychological need satisfaction, and parental burnout. Results showed that parents' ruminative exploration of identity was related to more parental burnout. Using a bifactor model for psychological need satisfaction, in which general need satisfaction was distinguished from the three specific needs (autonomy, competence, and relatedness), we found that autonomy satisfaction and general need satisfaction in particular were related to less parental burnout. Regarding the interplay of parental identity and psychological need satisfaction, we found systematic evidence for a moderating role, with the presence of one resource buffering the absence of the other resource. Overall, findings suggest that both parental identity and psychological need satisfaction serve as sources of resilience and may be interesting targets in prevention and intervention programs supporting parents at risk for parental burnout.

Keywords: Parental Burnout, Parental identity, Psychological Needs, Parenting, Bifactor Model

Highlights:

- This study examined parental identity and psychological need satisfaction as two resources protecting against parental burnout.
- For psychological need satisfaction, a bifactor approach was used, distinguishing a general need satisfaction factor from three specific needs.
- Whereas psychological need satisfaction was related to lower levels of parental burnout, ruminative exploration of identity was related to more parental burnout.
- We found systematic evidence for moderation, with the presence of one resource buffering the absence of the other resource.
- Both parental identity and parents' psychological need satisfaction could be interesting targets for prevention and counseling.

Becoming a parent is one of the most significant and potentially rewarding transitions in a person's life. It fundamentally changes one's way of living and one's identity as a person (Cowan & Cowan, 2000). However, parenting can also be very stressful and demanding (Crnic & Low, 2002). When the stress is too intense, too prolonged, and not compensated by the resources required to address it, parents may develop parental burnout (Mikolajczak & Roskam, 2018). Parents then experience exhaustion, feel unable to be the parent they want to be, and they ultimately even distance themselves from their children emotionally. In recent years, studies have attempted to gain a better understanding of parental burnout, its consequences and potential risk factors (Mikolajczak et al, in press). Relatively few studies, however, attempted to identify protective factors that reduce parents' vulnerability to burnout. Based on a cross-fertilization between the identity literature (Erikson, 1968; Kroger & Marcia, 2011) and self-determination theory (SDT; Ryan & Deci, 2000, 2017), this study aims to contribute to this research field by examining both the role of a clear parental identity, providing parents with a sense of direction, and parents' basic psychological need satisfaction, providing parents with energy resources, in protecting parents against burnout.

Parental Burnout

Although it has long been recognized that parenting is sometimes stressful, the theme of parenting stress has gained renewed attention since the observation that it can lead to parental burnout (Roskam et al., 2017; Roskam et al., 2021). Parental burnout is not the same as ordinary parenting stress (Brianda et al., 2020; Lebert-Charron et al., 2018) but occurs when parenting stress severely and chronically overwhelms parents' resources to cope (Mikolajczak et al., 2020). In other words, it is a consequence of a chronic imbalance between risks and resources in the parenting domain (Mikolajczak & Roskam, 2018). Specifically, parental burnout consists of four sets of symptoms, the most important one of which is emotional exhaustion (Mikolajczak et al., 2019; Roskam et al., 2018). Parents suffering from emotional exhaustion are overwhelmed by an extreme fatigue in their parental role and usually perceive that their parenting requires too much effort and energy in their daily activities and responsibilities. The second symptom is parents' emotional distancing from their children. Parents displaying this symptom gradually become less involved in the upbringing of their children. They function on autopilot and the interactions are limited to functional aspects and practical care (feeding, bringing the child(ren) to bed) without responding or being sensitive to the children's emotional needs. The third symptom is a loss of accomplishment and fulfillment as a parent. Parents feel fed up with parenting and become helpless in their role as a parent. They no longer enjoy being with their children and no longer experience any satisfaction or competence as parents. Finally, contrast is defined by the parents' awareness of the difference between the

parents they were and the parents they have become. They no longer recognize themselves, often resulting in feelings of guilt and shame (Hubert & Aujoulat, 2018; Roskam et al., 2018).

Parental burnout appears to affect a sizeable group of parents (Roskam et al., 2018; Roskam & Mikolajczak, 2020). According to a recent cross-cultural examination conducted by the International Investigation of Parental Burnout among thousands of parents in 42 countries around the world, 5 to 8% of parents are suffering from parental burnout, particularly in Western countries (Roskam et al., 2021). This prevalence rate is worrying because recent findings indicate that parental burnout has detrimental consequences for parents themselves and for their children as well (Mikolajczak et al., 2019).

Considering parents themselves or their mental health, parental burnout has been linked to escape ideations and severe suicidal thoughts (Mikolajczak et al., 2019), even to a stronger extent than job burnout or depression (Mikolajczak et al., 2020). In addition, parental burnout is also related to psychological forms of escape such as addictive behavior (Mikolajczak et al., 2018), sleep disturbances and health disorders (Mikolajczak et al., 2019; Van Bakel et al., 2018). At the biological level, parental burnout appears to increase cortisol levels (i.e., stress hormone) beyond the threshold above which it becomes detrimental to physical health (Brianda et al., 2020).

Parental burnout also affects children's development adversely through dysfunctional parenting practices. Parents experiencing high levels of burnout are at risk to become neglectful or even violent towards their offspring (Hubert & Aujoulat, 2018; Mikolajczak et al., 2018). Exhausted parents more often engage in harsh parenting (Hansotte et al., 2021). These findings have also been documented longitudinally, with parental burnout predicting increases across time in child neglect and violent parental behaviors towards children (Mikolajczak et al., 2019).

Given the detrimental consequences of parental burnout for parents' mental health and for children's development, it is essential to identify resources that potentially protect parents against this condition. Unfortunately, most research on the antecedents of parental burnout has focused rather uniquely on risk factors. Mikolajczak et al. (2017) categorized the risk factors for parental burnout into five types: socio-demographics (e.g., having a large family, Kawamoto et al., 2018), child characteristics (e.g., having a child with special needs, Gérain & Zech, 2018), parents' stable traits (e.g., perfectionism, Kawamoto et al., 2018), parenting cognitions and behaviors (e.g., a lack of emotion and stress-management abilities, Mikolajczak et al., 2018), and family functioning (e.g., low cooperation between co-parents, Gérain & Zech, 2018). Although many risk factors have already been identified, effective prevention and intervention is best aimed at strengthening resources and

protective factors, rather than merely reducing risk factors. To date, only a few studies have focused on protective factors. For instance, a recent meta-analysis showed that cognitive reappraisal, an adaptive emotion regulation strategy, related negatively to parental burnout (Brandão et al., 2023). In addition, cognitive reappraisal buffered the negative impact of parental stress (Vertsberger et al., 2021) and a lack of social support on parental burnout (Lin et al., 2022a). As another example, emotional intelligence was shown to relate negatively to parental burnout and buffered the negative impact of perfectionism on parental burnout (Lin et al., 2022b; Lin & Szczygieł, 2022). Based on identity literature (Erikson, 1968) and SDT (Ryan & Deci, 2017), we propose two additional resources against parental burnout that respectively provide purpose to the parental role (i.e., parental identity) and the necessary energy for parents (i.e., psychological need satisfaction) to realize their parental goals.

Parental Identity as a Resource for Parental Functioning

The transition to parenthood is essentially an identity-changing episode as it adds a new dimension to an individual's sense of self and requires an adjustment of one's personal life and a restructuring of goals, behaviors, and responsibilities (Mercer, 2004). According to Erikson (1968), identity formation is a crucial part of individuals' psychosocial development. To develop a clear personal identity, individuals need to answer crucial existential questions about who they are, their most important values, and their future goals. Erikson (1959) considered identity development as a critical developmental task in adolescence, but also emphasized strongly that identity development is a continuous process throughout the lifespan (Kroger, 2015). As adults, parents also need to develop and integrate their parental identity, that is, the degree to which they have clear and coherent commitments in their parental role (Fadjukoff et al., 2016). These commitments represent individuals' beliefs and attitudes about their role as a caregiver, including the processes through which these beliefs and attitudes are developed (Cast, 2004).

Abundant research among adolescents and emerging adults has consistently documented that identity plays an important role in psychological adjustment. In this literature, mature identity development is characterized in terms of two crucial processes, namely commitment and exploration (Marcia, 1966). Whereas commitment involves the formulation of clear values, and beliefs that define who one is, exploration involves a thorough reflection about possible commitments and a systematic weighing of alternatives. A more mature identity is related to better personal well-being and psychosocial functioning (Meeus, 2011; Schwartz et al., 2011). Commitment, in particular, has been found to be associated with emotional security and positive mental health, with associations between exploration and mental health being less straightforward (Lindekilde, 2018). In

contrast, some individuals experience difficulties in forming a stable and meaningful identity, marked by existential doubts or even alienation. These individuals are at risk for mental problems (Klimstra & Denissen, 2017). To capture this difficult identity exploration process, Luyckx et al., (2008) proposed the inclusion of ruminative exploration as a maladaptive dimension. Ruminative exploration reflects profound doubts about one's commitments and a tendency to brood endlessly about identity-relevant choices (Luyckx et al., 2008). Ruminative exploration has been associated with internalizing symptoms and a fragmented sense of self (Crocetti et al., 2007; Luyckx et al., 2008).

Surprisingly, given that being a mother or father is a central and important aspect of parents' identity (Piotrowski, 2021), research on parental identity is rather scant. Only a few studies have examined dimensions of identity development in the context of parenthood and have begun to demonstrate that parental identity is related to parents' well-being. Using semi-structured interviews and a longitudinal design, Fadjukoff et al. (2016) found that both mothers and fathers developed towards more mature levels of parental identity as they grew older. Importantly, a more achieved parental identity was related to better quality of parenting as well as to higher parental well-being (with the latter association emerging mainly among fathers). Piotrowski (2018) adapted a questionnaire developed for research on adolescent identity formation (Crocetti et al., 2007) to the topic of parental identity and found that parental identity commitment was associated systematically with positive mental health outcomes such as life satisfaction, whereas reconsideration of commitments (i.e., uncertainty and instability of commitment) was associated with negative mental health outcomes such as anxiety. Similarly, Meca et al. (2020) found evidence for a link between clear parental identity commitments and reduced risk for internalizing problems among mothers and fathers. Even in uncertain times such as the COVID-19 pandemic, [Schrooyen, et al., 2021] showed that parents with a clearer parental identity reported less negative parental experiences and ill-being and more positive parental experiences and well-being.

These findings underscore the role of parental identity in parents' adaptation and suggest that parental identity processes may also play a protective role against parental burnout. Parents with clear and well-explored identity commitments would have a strong sense of direction that helps them overcome the inevitable hassles and challenges inherent to parenthood. Being aware of the place their parental role serves in their identity and having explored the values their role serves in their functioning, parents may be better able to put problems into perspective and to maintain a sense of purpose, even during difficult parenting episodes or hassles. In contrast, parents with a high level of ruminative exploration in their parental identity tend to be more susceptible to frustrations or stress associated with parenting hassles, and, therefore, when these parents encounter parenting

stresses, they quickly doubt their capacity for parenting and devalue their parental role for their identity. These parents begin to feel incompetent in their parenting role. The most direct initial evidence for the role of parental identity in parental burnout was provided by Piotrowski (2021), who showed that parents with higher levels of difficulties in forming a stable sense of parental identity (i.e., with higher reconsiderations of their commitment or with more ruminative exploration) also reported higher levels of parental burnout. In a recent follow-up of parents during the COVID-19 crisis, Piotrowski (2022) showed that parents with a more mature parental identity (high commitment and low ruminative exploration) were less likely to be in a developmental trajectory characterized by high or increasing levels of parental burnout. Because only two studies to date examined associations between parental identity and parenting burnout (Piotrowski, 2021, 2022), replication and extension of these findings is needed. Therefore, the present study sought to examine the associations between three dimensions of parental identity and parental burnout. More specifically, we sought to investigate whether levels of parental identity commitment (i.e., degree to which individuals have committed to a parenting identity), parental identity exploration (i.e., degree to which individuals have explored what parenting means to them), and parental identity ruminative exploration (i.e., degree to which individuals have pervasive doubts about their parental role) were significantly associated with parents' perceptions of their level of parental burnout.

Parents' Psychological Need Satisfaction as a Resource for Energy

In addition to having a clear parental identity, which can serve as a stable anchor buffering against the development of parental burnout, parents also need sufficient energy to invest in the parental role. Indeed, parents ideally have both a clear set of goals as well as energy to pursue and realize these goals. According to SDT, three psychological needs are crucially important for individuals' well-being, psychological growth and mental energy (Deci & Ryan, 2000; Ryan & Deci, 2017; Vansteenkiste et al., 2020). Specifically, this theory focuses on the needs for autonomy (a sense of volition and psychological freedom in one's activities), relatedness (the experience of reciprocal care and close connection with important others), and competence (a sense of mastery and effectiveness), assuming that these needs are universal and fundamental for psychological health. Ample literature has confirmed that psychological need satisfaction indeed plays a major role in mental health (for reviews, see Ryan & Deci, 2017; Vansteenkiste et al., 2020).

Recent developmental research based on SDT has begun to show that psychological need satisfaction also matters for parents' adaptation and for the quality of their parenting. The basic assumption in this research is that when parents' own needs are satisfied, parents would feel more vital and energetic, and be better able to be attuned to their children's needs, whereas they would experience more distress, become vulnerable to

psychopathology, and risk engaging in more dysfunctional parenting, when their needs are frustrated (Mabbe et al., 2018). Consistent with this assumption, several studies have documented the beneficial effects of parental psychological need satisfaction on the well-being and vitality of parents. Van der Kaap-Deeder et al. (2019) showed that parents' psychological need satisfaction was related to more psychological availability and lower stress in parent-child interactions. Brenning et al. (2015) found evidence for an association between prenatal psychological need satisfaction and mothers' prenatal well-being, while Gauthier et al. (2010) reported evidence for a similar association postnatally. In the absence of psychological need satisfaction, parents become more quickly stressed out by the daily hassles of parenting. Furthermore, studies have shown that need-based experiences are a critical resource for parenting, with parents' psychological need satisfaction being significantly associated with more supportive parenting practices (van der Kaap-Deeder et al., 2019). Several diary studies have demonstrated that parents' daily psychological need satisfaction was related positively to daily autonomy-supportive parenting and negatively to psychologically controlling parenting (Dieleman et al., 2018; Mabbe et al., 2018).

On the basis of SDT and previous research, we argue that the energizing role of psychological need satisfaction can counteract perceptions of parental burnout. As parental burnout arises when there is a chronic lack of resources to meet parenting needs, leading to a depletion of energy and emotional exhaustion, psychological need satisfaction can foster parents' energy level and recharge their batteries, such that parents are better protected against the risks of burnout. In the context of job burnout, there is already considerable evidence that employees with higher levels of psychological need satisfaction are less prone to developing burnout (Trepanier et al., 2013; Van den Broeck et al., 2008). To date, similar effects in the field of parenting have not yet been demonstrated.

To provide a direct and detailed examination of the role of psychological need satisfaction in burnout, we adopt a novel analytical approach (i.e., bifactor modelling) distinguishing between parents' general psychological need satisfaction and their unique satisfaction of the three specific needs (Chen et al., 2012; Reise, 2012). Doing so is important, because specific psychological needs may play a particularly prominent role in certain domains of parenting. Specifically, addressing the domain of autonomy in parenting, Schrooyen et al. (2021) showed that a parental need for autonomy was satisfied the least during the COVID-19 period (probably because of the restrictions parents experienced and the difficulty combining work with the parental role) and had the strongest association with parental mental health. As such, research examining only the role of general need satisfaction can miss the important unique associations with specific needs. Ideally, research considers both

parents' general level of need satisfaction as well as the specific effects of the three needs separately (Chen et al., 2015). The bifactor model is an ideally suitable analytical approach in this regard. Several SDT-based studies, examining the associations between psychological need satisfaction and mental health, have demonstrated the merits of this approach (Gillet et al., 2019; Sánchez-Oliva et al., 2017), but no study to date applied the bifactor approach to research on parental adaptation.

Interplay between Parental Identity and Parents' Psychological Need Satisfaction

Because parents' identity and their psychological need satisfaction can both be considered as important resources arming parents against burnout, it is important to also examine their interactive interplay. Possibly, these two resources have synergistic effects on parental burnout. In that case, parents' identity and their psychological need satisfaction could both become uniquely important as well as reinforce each other's effects. If it is indeed important for parents to have both a sense of direction and sufficient energy to realize one's goals as a parent, then both resources could be ideally combined and may even strengthen each other's effects. Technically, such a synergistic effect comes down to a moderating effect, where parental identity and psychological need satisfaction interact positively in the prediction of parental burnout. To the best of our knowledge, no study to date tested the possibility of moderating effects of parental identity and psychological need satisfaction on parental burnout.

The Present Study

The aim of this study is to examine the unique and interactive roles of parental identity and parents' psychological need satisfaction in parental burnout. This overall aim encompassed three more specific hypotheses. First, based on the identity literature, we expected that high levels of parental commitment and exploration will be associated with lower levels of parental burnout, and that high levels of parental ruminative exploration will be associated with higher levels of parental burnout (Hypothesis 1). Second, on the basis of SDT and using a bifactor analytical approach distinguishing between parents' general need satisfaction and satisfaction of the three specific needs, we expected that general need satisfaction (and perhaps specific need satisfaction) would relate negatively to parental burnout (Hypothesis 2). Finally, we address the interplay between parental identity and parents' need satisfaction, thereby hypothesizing that parental identity and parents' need satisfaction have synergistic effects on parental burnout (moderation). The combined effects of parental identity and parents' need satisfaction would then go beyond their individual effects (Hypothesis 3).

To examine the unique and interactive roles of parental identity and parents' psychological need satisfaction in parental burnout, over and above the role of sociodemographic risk factors, we controlled

statistically for relevant sociodemographic risk factors. Doing so is deemed important because previous studies have identified several sociodemographic risk factors for parental burnout. For instance, studies have shown that being a woman (Mikolajczak & Roskam, 2018), having a large family (Kawamoto et al., 2018), having young children at home (Le Vigouroux & Scola, 2018; Mikolajczak et al., 2017), having a lower educational level (Hatta & Kpassagou, 2023; Woine et al., 2021) and working part-time or being a stay at-home parent (Lebert-Charron et al., 2018) increase the risk for parental burnout.

Method

Study Design and Procedure

The design of the current study is cross-sectional and correlational. The study was conducted as part of the International Investigation of Parental Burnout, which is a worldwide consortium involving 42 countries coordinated at UC Louvain in Belgium (Roskam et al., 2021). We disseminated the survey through various platforms, including our professional network on Facebook, specific online forums dedicated to parenting (e.g., mothers' discussion groups), organizations specializing in parental support, and parenting blogs. Special efforts were made to include blogs for fathers to ensure that we would recruit both mothers and fathers. Parents were invited to participate if they had at least one child living at home. This criterion was stated clearly in the invitation to ensure that only relevant participants were recruited. Before beginning the survey, parents were required to give informed consent. They were informed that participation was voluntary, that they could withdraw at any time without justification, and that their responses would remain anonymous because no personal identifying information was collected. After clicking on the survey link, participants were asked preliminary questions to confirm that they met the eligibility criteria (i.e., having at least one child living at home). To ensure the quality of the responses and to filter out non-attentive participants, we included attention check questions such as, “please choose number 4 here.” This method is a commonly recommended practice in online surveys to confirm that respondents are reading the questions carefully and are human participants, not automated bots (Abbey & Meloy, 2017). Of the 748 people who clicked on the survey link, 476 continued beyond the initial questions. After inspection of the attention the check questions, 26 participants were removed for failing to respond correctly, leading to a final sample of 450 parents. The Ethical Committee of Ghent university approved this study protocol (nr. 2018/20). Data were collected between the beginning of April to the end of June 2018.

Participants' Characteristics

Of the 450 respondents (97.1% born in Belgium), 287 (63.8%) were women and 163 (36.2%) were men. All respondents self-identified as parents and reported living with at least one child in the same household.

Participants' age ranged from 23 to 58 years ($M = 35.48$, $SD = 7.37$) for mothers and from 26 to 59 years ($M = 38.39$, $SD = 7.58$) for fathers. Overall, the number of children in the participating families ranged from 1 to 5 ($M = 1.86$, $SD = 0.83$) with the majority of parents having two children (46.2%), followed by one child (36.4%) and three children (12.7%). Only a small percentage of parents had four (4.2%) or five children (0.4%). The ages of these children ranged from 0 to 28 years ($M = 4.32$, $SD = 5.18$) for the youngest and from 0 to 34 years ($M = 6.91$, $SD = 6.50$) for the oldest (or only) child. When parents reported that the child is 0 years old, it indicates that they have an infant younger than 12 months. Of the parents, 85.1% lived in an intact family with both parents present, 8.2% reported being a single-parent, and 4.4% lived in a reconstituted family. The remaining 2.3% of the parents reported having another family structure (e.g., foster or adoptive family). Based on the respondents' subjective perceptions of the neighborhood conditions in which they lived, 56.7% perceived their neighborhood as average, 40.4% considered their neighborhood relatively prosperous, and 2.9% viewed their neighborhood as relatively disadvantaged. Regarding the educational level of the parents, 0.9% obtained a primary education degree, which is equivalent to completing elementary school in the U.S. (grades 1-5 or 1-6), 17.5% obtained a secondary education degree, equivalent to graduating from high school in the U.S. (grades 9-12 or 10-12), 43.6% obtained a college degree (non-university higher education at a college), similar to obtaining an associate's degree or a bachelor's degree from a community college in the U.S. and 38% obtained a university degree, which is comparable in the U.S. to obtaining a bachelor's, master's, or doctoral degree from a university. With regard to their work situation, 73.6% of the parents worked fulltime, while 19.3% worked part-time. The remaining 7.1% did not work (4.7% due to unemployment or work incapacity, and 2.4% were stay-at-home parents).

Measures

Parental Burnout. The Parental Burnout Assessment (PBA, Roskam et al., 2018), translated to Dutch by the authors, was used to assess parental burnout. The internal structure of the Dutch version of the PBA has been tested and demonstrated by Roskam et al. (2021) through a CFA. In our study, we also conducted a CFA which yielded an acceptable fit ($CFI = .99$, $RMSEA = .08$, $SRMR = .01$, $\chi^2(2) = 7.67$; $p < 0.05$). The scale comprises 23 items assessing the four core symptoms of parental burnout: Emotional Exhaustion (9 items, e.g., *I feel completely run down by my role as a parent*), Emotional Distancing from one's children (3 items, e.g., *I do what I'm supposed to do for my children but nothing more*), Feelings of Being Fed Up (5 items, e.g., *I don't enjoy being with my children*), and Contrast with Previous Parental Self (6 items, e.g., *I don't think I'm the good father/mother that I used to be*). All items are rated on a 7-point Likert scale (1 = *never*, 2 = *a few times a year or*

less, 3 = *once a month or less*, 4 = *a few times a month*, 5 = *once a week*, 6 = *a few times a week*, 7 = *every day*).

All subscales showed a good internal consistency with Cronbach's alpha 0.95, 0.83, 0.89 and 0.93 respectively.

To compute a composite score for parental burnout, we averaged the scores from the four subscales, each addressing one of the core symptom areas. Higher scores on this total scale indicate a greater presence of parental burnout symptoms. The reliability of this total scale was found to be high, with a Cronbach's alpha coefficient of 0.95. To compute the clinical cut-off scores for parental burnout, in line with the procedure of Brianda et al. (2023), a total score was computed by summing the four subscales. Each participant with a sum score higher than 109.3, can be identified as suffering from parental burnout.

Parental Identity. The measurement of parental identity was based on the Dutch version (Luyckx et al., 2008) of the Dimensions of Identity Development Scale (DIDS; Luyckx et al., 2008), and adapted to the parental role. For instance, the item 'I Know what I want to achieve in my life' was adapted to 'I know what I want to achieve as a parent'. This adapted measure was used successfully in previous research with Dutch parents (Schrooyen et al., 2021). Using our own data, we conducted a CFA, which yielded an acceptable fit (CFI = .88, RMSEA = .08, SRMR = .10, Chi Square (84) = 301.68 $p < .001$). Most importantly, all factor loadings had a p -value $< .001$ (see Table 1 in the Supplementary Files). We used the items of three identity scales: Commitment (4 items, e.g., *I know what I want to achieve as a parent*), Exploration (4 items, e.g., *I think actively about different ways to fill in parenthood*), and Ruminative exploration (4 items, e.g., *I worry about what I should do as a parent*). All items were rated on a 5-point Likert scale ranging from 1 (*completely disagree*) to 5 (*completely agree*). Cronbach's alpha reliabilities for the scales were 0.82, 0.81 and 0.79, respectively.

Psychological Needs. To measure parents' satisfaction of the psychological needs for autonomy, competence, and relatedness, we used a Dutch version (Chen et al., 2015) of the Basic Psychological Need Satisfaction and Need Frustration Scale (BPNSNFS, Chen et al., 2015). Although the original scale consists of 24 items, a short 12-item version was used, which has been validated and used successfully in previous research with a sample of Dutch parents (Mabbe et al., 2018). The three psychological needs (autonomy, competence, and relatedness) were each measured by 4 items, two of which measured satisfaction of the need and two of which measured frustration of the need: Autonomy Satisfaction (e.g., *I feel a sense of choice and freedom in the things I undertake*) and Frustration (e.g., *I feel forced to do many things I wouldn't choose to do*), Competence Satisfaction (e.g., *I feel confident that I can do things well*) and Frustration (e.g., *I feel insecure about my abilities*), and Relatedness Satisfaction (e.g., *I feel connected with people who care for me, and for whom I care*) and Frustration (e.g., *I feel excluded from the group I want to belong to*). All items were rated on a 5-point Likert

scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). In line with previous research (e.g., Chen et al., 2015), the need frustration items were reverse coded and a composite score was calculated for each need (i.e., autonomy, relatedness, and competence) by averaging the satisfaction items with the reverse coded frustration items. This approach yielded good internal consistencies (autonomy satisfaction: $\alpha = 0.77$, competence satisfaction: $\alpha = 0.76$, and relatedness satisfaction: $\alpha = 0.75$). Cronbach's alpha of the total satisfaction score was 0.87.

Plan of Analysis

To test our hypotheses, we relied on structural equation modeling (SEM) with latent variables. We used the robust maximum likelihood (MLR) estimator available in Mplus 8.8 (Muthen & Muthen, 1998-2017), which provides standard errors and goodness-of-fit indices that are robust to the non-normality of the data. Because Little's test (Little, 1988) revealed that missing data were missing completely at random, $\chi^2(2844) = 344.55$, $p = 1.00$, we used full information maximum likelihood estimation to handle missing data (Schafer & Graham, 2002).

Prior to estimating the structural equation models testing the hypotheses, confirmatory factor analysis (CFA) was used to determine the extent to which the observed variables measured the latent constructs. Herein, the items of the Dimensions of Identity Development Scale were used as indicators of the three identity constructs, whereas the four subscales of the Parental Burnout Assessment were used as indicators of parental burnout. Because the three needs were highly correlated, which may result in biased prediction of the dependent variable (parental burnout), we used the bifactor model, distinguishing a general need satisfaction factor from three specific (and uncorrelated) need satisfaction factors (DeMars (2013); Reise, 2012). In this bifactor model, all latent variables are uncorrelated. As a consequence, the General Need Satisfaction factor captures all variance that is shared by the three separate psychological needs (for autonomy, competence, and relatedness). This is important, as the specific psychological needs as well as general need satisfaction may all play a particularly prominent role in the domain of parental burnout. Therefore, the bifactor model is an ideally suitable analytical approach as it considers both parents' general level of need satisfaction as well as the specific effects of the three needs separately (Chen et al., 2015). To examine the main effects of parental identity (hypothesis 1), a model with latent variables was examined in which the three identity constructs (i.e., commitment, exploration and ruminative exploration) were added as predictors of parental burnout. To examine the main effects of psychological need satisfaction (hypothesis 2), a model with latent variables was examined in which general need satisfaction and specific need satisfaction of autonomy, relatedness and competence were added as

predictors of parental burnout. To test the moderation hypothesis (hypothesis 3), three models with latent variable interactions were examined, one with commitment, one with exploration and one with ruminative exploration as a predictor. These models included four possible interactions with the three specific need satisfaction factors and the general need satisfaction factor as moderators.

In all models, we controlled for significant socio-demographic variables. Model fit was evaluated by several fit indices: the normed χ^2 test (acceptable when χ^2/df ratio is 2 or below), the Comparative Fit Index (CFI; minimal threshold of 0.95), the Standardized Root Mean square Residual (SRMR; maximum threshold of 0.08), and the Root Mean Square Error of Approximation (RMSEA; maximum threshold of 0.06) (Hu & Bentler, 1999; Kline, 2005).

Results

Descriptive Statistics and Preliminary Analyses

Descriptive statistics and the correlation matrix for all study variables are presented in Table 1. Based on the clinical cut-off scores for parental burnout provided by Brianda et al. (2023), 5.6% of the parents in this study suffered from clinically elevated levels of parental burnout. In the bivariate analysis of parental burnout, findings show that fathers (gender code = 1) had significantly ($p < 0.001$) lower parental burnout than mothers, as the correlation between gender and parental burnout is an inverse relationship. Parents' education level was not significantly ($p > 0.05$) related to parental burnout. For parental identity, lower commitment was significantly ($p < 0.001$) correlated with higher parental burnout, whereas higher parental identity exploration and higher ruminative exploration were significantly ($p < 0.05$ and $p < 0.001$, respectively) correlated with higher parental burnout. For psychological need satisfaction, lower scores on total need satisfaction, and on the individual needs of autonomy, relatedness, and competence, were all significantly ($p < 0.001$) associated with higher scores on parental burnout.

[Insert *Table 1* here]

To determine whether demographic variables were associated with our study variables, we performed a multivariate analysis of variance (MANCOVA), with commitment, exploration, ruminative exploration, satisfaction of autonomy, relatedness, and competence, and parental burnout as the dependent variables, and with gender, age, educational level, family structure, work situation, number of children, and age of the oldest child as independent variables. We found a multivariate (Wilks' Lambda) significant effect of gender ($F(7,434) = 6.86, p < .001; \eta^2 = 0.10$), and educational level ($F(7,434) = 5.22, p < .001; \eta^2 = 0.08$). The results are presented in Table 1. Based on these results, we decided to control for parents' gender and educational level in all further analyses.

Next, we estimated a bifactor CFA model for need satisfaction (see Figure 1), in which all items were allowed to define one general need satisfaction factor and in which the items for autonomy, relatedness, and competence defined the corresponding three specific factors. To identify this model, the variance of each latent factor was set to 1. All of these factors were specified to be uncorrelated, thereby ensuring that relationships between the three specific factors were captured completely in the general need satisfaction factor (Brunet et al., 2016; Reise et al., 2007; Reise, 2012).

[Insert *Figure 1* here]

We then inspected the fit of the complete measurement model, comprising the independent variables (commitment, exploration and ruminative exploration, general need satisfaction, autonomy satisfaction, relatedness satisfaction, competence satisfaction), and the dependent variable (parental burnout). The latent constructs of commitment, exploration and ruminative exploration were represented by their respective items, while parental burnout was modelled by the four subscales. The measurement model yielded a good fit: $X^2(313) = 674.48$, $p < 0.001$; CFI = 0.92, RMSEA = 0.05; SRMR = 0.06; $X^2/df < 5$ (Ahmad et al., 2016). A reliable measurement model was thus obtained, indicating that it was appropriate to test the structural models (see Table 1 in the Supplementary Files).

Parental Identity and Parental Burnout (Hypothesis 1)

First, we tested the main effects of commitment, exploration and ruminative exploration on parental burnout (see Supplementary Figure 1). This model had an acceptable fit: $X^2(122) = 409.27$, $p < 0.001$; CFI = 0.91, RMSEA = 0.07; SRMR = 0.07; $X^2/df < 5$ (Ahmad et al., 2016). As expected, ruminative exploration predicted higher levels of parental burnout ($\beta = 0.65$; $p < 0.001$), whereas commitment and exploration were unrelated to parental burnout. Ruminative exploration was negatively associated with commitment ($\beta = -0.23$; $p = 0.01$), and positively with exploration ($\beta = 0.35$; $p < 0.001$). Commitment and exploration were positively associated ($\beta = 0.71$; $p < 0.001$).

Parents' General and Specific Psychological Need Satisfaction and Parental Burnout (Hypothesis 2)

Second, we examined whether parents' psychological need satisfaction, both specific and general related to parental burnout (see Supplementary Figure 2). This model had an acceptable fit: $X^2(118) = 336.32$, $p < 0.001$; CFI = 0.93, RMSEA = 0.06; SRMR = 0.06; $X^2/df < 5$ (Ahmad et al., 2016). As expected, parental burnout related negatively to general need satisfaction ($\beta = -0.52$; $p < 0.001$). With respect to the specific factors, parental burnout related negatively to autonomy satisfaction ($\beta = -0.72$; $p < 0.001$) and to relatedness satisfaction ($\beta = -0.28$; $p < 0.001$).

The Moderating Role of Parents' General and Specific Psychological Need Satisfaction (Hypothesis 3)

Next, we tested three moderation models, one for each parental identity dimension. Each model included the four possible interactions with the general need satisfaction factor and the three specific need satisfaction factors in the prediction of parental burnout. We found evidence for eight moderating effects (out of twelve tested effects). For an overview of the interaction effects, see Table 2, and see Table 3 for the gradients of simple slopes.

[Insert *Table 2 and Table 3* here]

Regarding commitment, all four possible interactions with need satisfaction were significant. Interestingly, all interaction effects showed the same interaction pattern. Specifically, high levels of need satisfaction seemed to buffer the negative association between low commitment and parental burnout. For instance, as shown in Figure 2, scores for parental burnout were low when commitment was high (independent of general need satisfaction) and when commitment was low but general need satisfaction was high. Conversely, scores for parental burnout were most pronounced when parents simultaneously reported low commitment and low general need satisfaction, suggesting that low levels of both commitment and need satisfaction displayed synergistic associations with parental burnout. The same interaction pattern applied to the specific need satisfaction factors. Specifically, the association between commitment and parental burnout was more pronounced when autonomy, relatedness, or competence satisfaction was low, and less pronounced when autonomy, relatedness, or competence satisfaction was high, suggesting a buffering role of need satisfaction.

As expected, the main effect for ruminative exploration had a positive relationship with parental burnout; meaning that higher scores for ruminative exploration were associated with higher scores for parental burnout. As also expected, the main effects for specific psychological needs had negative (inverse) relationships to parental burnout. Of the psychological needs, there was no significant interaction between competence satisfaction and ruminative exploration in relation to parental burnout. For the general need satisfaction, autonomy satisfaction, and relatedness satisfaction, these interactions with ruminative exploration were significant. Specifically, when parents scored high on ruminative exploration, high levels of general need satisfaction, or autonomy satisfaction, or relatedness satisfaction, all played a buffering role in the prediction of parental burnout. For example, as can be seen in Figure 3, the positive association between ruminative exploration and parental burnout was more pronounced when general need satisfaction was low, and less pronounced when general need satisfaction was high.

Parental identity exploration had no significant independent main effect or any interactive (moderating) effect on parental burnout, as can be seen in Table 2 and Table 3; although, there was a small positive relationship with parental burnout, originally detected with bivariate correlation, in Table 1. In the context of parental identity exploration, psychological need satisfaction (general, autonomy, relatedness, and competence) were more important to parental burnout. Specifically, scores for parental burnout were most pronounced when parental identity exploration was low, while simultaneously parental scores were also low on any of their specific need satisfaction, thus creating a potential for parental burnout due to low psychological need satisfaction. At high levels of any of the parental psychological need satisfaction, there was no significant difference in parental burnout as a function of exploration (see Figure 4).

Overall, we can conclude that all interaction effects found showed a similar pattern, with associations between identity and parental burnout being more pronounced at low levels of need satisfaction and less pronounced at high levels of need satisfaction, suggesting that need satisfaction plays a buffering role.

[Insert Figure 2, Figure 3 and Figure 4 here]

Discussion

Given the important implications of parents' well-being for their quality of parenting and their children's development, it is critical to identify protective factors for parental burnout, a key indicator of parental ill-being. Based on identity literature (Erikson's, 1968; Fadjukoff et al., 2016) and SDT (Ryan & Deci, 2017), this study advanced parental identity and psychological need satisfaction as two resources, potentially protecting parents against parental burnout. In the following sections, we discuss the associations between the parental identity dimensions, psychological need satisfaction, and parental burnout, and also the possible implication of these associations for intervention practices and future research.

Parents Need Both a Sense of Direction and Resources for Energy

Before examining the associations of parents' identity and need-based experiences with parental burnout, we inspected the mean level scores of these variables (Table 1). In the current sample, parenthood was experienced by the participants as predominantly positive, with parents generally reporting high scores on identity commitment, exploration and psychological need satisfaction, and low scores on ruminative exploration and parental burnout. This is not to say that parental burnout was entirely absent, because 5.6% of the participants reported clinically elevated levels of parental burnout (defined by a sum score higher than 109.3; Brianda et al., 2023). Another important nuance is that not all psychological needs were equally satisfied. Parents perceived their need for autonomy as less satisfied compared to the needs for relatedness and

competence. This finding is in line with the results of Schrooyen et al. (2021) and consistent with the assumption that parenthood comes with many responsibilities and obligations that limit parents' choices and psychological freedom. A final nuance is that mothers reported more negative (parental) experiences than fathers (Table 1), a finding consistent with many previous studies (Nelson-Coffey et al., 2019), and indicating that mothers today still invest more strongly in the parental role and experience the burdens of this role more strongly than fathers. Overall, the findings from this study do not confirm the sometimes one-sided and bleak portrayal of contemporary parenthood in Western societies as a psychological burden (Dupont et al., 2022). Most parents fare well in the parenting role, thereby avoiding the pitfalls of exhaustion, incompetence, and emotional distancing. Of course, there are vast individual differences between parents in these experiences, an observation raising the question as to which parents are better protected against parental burnout.

As expected and consistent with previous research by Piotrowski (2018, 2022), parents high on ruminative exploration reported higher levels of parental burnout. Parents with pervasive doubts, who lack purpose and direction in their parental role (ruminative exploration), become more easily exhausted as parents. With respect to commitment, the findings were somewhat inconsistent. Commitment was unrelated to parental burnout when controlling for the other two identity dimensions, yet related negatively to parental burnout when examining it in isolation from the other two identity dimensions and over and above the effect of parents' psychological need satisfaction. This latter finding is consistent with our reasoning that having a clear view of what kind of parent they are and what they stand for probably provides parents with guidance and confidence, making them less prone to developing parental burnout. However, this inconsistency in findings suggests that the direct beneficial role of commitment in parental burnout is less robust compared to the detrimental role of ruminative exploration. Finally, no significant associations were found between parental identity for exploration and parental burnout in the present study. Possibly, once being a parent, the degree of parental identity exploration seems less pivotal, unless it is characterized by hesitation, indecisiveness and doubt, portraying ruminative exploration. However, maybe for first-time parents and in the early phases of parenthood exploration might be more important (Seppälä et al., 2022). Also, parental identity commitment and exploration could be more important for outcomes other than parental burnout, such as parenting quality, an interesting avenue that could be explored in future research.

In addition to having a clear parental identity, parents need sufficient energy (Mabbe et al., 2018; Nelson et al., 2014; van der Kaap-Deeder et al., 2019). Indeed, our results showed that having more satisfaction of the psychological needs for autonomy, relatedness and competence is related to lower levels of parental

burnout. Using the innovative bifactor approach, which we used for the first time in the context of parenthood experiences, we were able to distinguish between the role of parents' general need satisfaction and the unique roles of satisfaction of the three specific needs in the prediction of parental burnout. However, we observed that the factor loadings for some items on the unique need satisfaction factors were relatively low (see Supplementary Table 1), indicating that these factors may be estimated with less reliability. One potential reason for this could be the use of a shortened version of the BPNSNFS. To address this limitation, we suggest that future research employ the full version which may yield stronger factor loadings and enhance the reliability of these unique factors. In line with earlier studies (Schrooyen et al, 2021) autonomy satisfaction was found to be most important. Experiencing a sense of authenticity and joy in life, albeit not specifically in the domain of parenting, may be the most protective factor against parental burnout. Future research could distinguish between autonomy satisfaction in the domain of parenthood and autonomy satisfaction in life domains other than parenthood. Although the link between experiencing autonomy in the parental role and parental burnout is more obvious, experiencing autonomy in other life domains, such as at work, may provide the necessary oxygen to deal with parent-related stress and worries. In addition, general need satisfaction was found to be important, a finding consistent with the assumption that the three needs mutually reinforce one another. Given the intricate associations among the three needs, it makes sense that what they have in common (i.e., the variance captured in the general factor) is quite strongly predictive for parental burnout. Also the need for relatedness still showed a specific and unique association with parental burnout, underscoring the importance of a strong and supportive social network, including a high-quality partner relationship. Overall, our results lend support to the assumption that satisfaction of parents' own psychological needs contributes to their vitality and well-being (Ryan & Deci, 2008). The observation that the need for competence was not uniquely associated with parental burnout, when examining possible moderation by ruminative exploration, does not mean that competence satisfaction is not important. Indeed, the competence satisfaction contributed strongly to the general need satisfaction factor. Moreover, competence might be more important in other specific life periods and social conditions and for other outcomes, such as parenting quality.

These findings regarding parents' need-based experiences again have practical implications, emphasizing the importance of parents being aware of and attentive to the satisfaction of their own psychological needs. To avoid that parents reach the point where they completely lack the energy to handle the daily responsibilities and challenges of parenthood, experiences of psychological need satisfaction are key to the perception of their parenting role (van der Kaap-Deeder et al., 2015, 2019). Intervention programs could include

‘need crafting’ exercises, wherein parents are encouraged to seek and create need-satisfying conditions and provide strategies to avoid energy-depleting experiences in order to recharge their batteries (Laporte et al., 2021). Future research could examine whether need crafting is indeed beneficial for parents’ resilience and may protect them from developing parental burnout.

The Interplay Between the Psychological Resources of Parental Burnout

Next, a central goal of this study was to investigate the interactive interplay between parental identity and parents’ psychological need satisfaction in parental burnout. Overall, we found consistent evidence for an interactive interplay between parental identity and psychological need satisfaction. In all identified interactions, the presence of an adaptive parental identity dimension (i.e., high on commitment, low on ruminative exploration, or low on exploration), or high psychological need satisfaction resulted in lower parental burnout scores. In other words, when one of these resources are present, which means that parents have a clear view of their parental role, without serious doubts, or they experience a good amount of psychological need satisfaction in their lives, they are well protected against feelings of exhaustion as a parent. Thus, the presence of one of the two resources compensates for the absence of the other resource. But when parental identity resources were lacking, (i.e., with scores low on commitment or high on ruminative exploration or high on exploration), and parents also experienced a low amount of psychological need satisfaction in their lives, then, parents become significantly more likely to be at risk for parental burnout. When confronted with stressors, these parents have no fallback options for resilience available, rendering them highly vulnerable to burnout. Especially this group of parents may benefit from interventions aimed at strengthening and replenishing their mental resources.

The results regarding the interaction effects reveal that parental identity commitments and exploration can play the anticipated protective role against parental burnout. These findings indicate that these identity dimensions may matter when needed most: when parents (temporarily) experience low need satisfaction, it helps to have a clear and well-explored identity that provides guidance for the future. From an applied point of view, our findings regarding parental identity underscore the importance for parents to reflect thoroughly about the centrality of the parental role in their overall identity and to be aware of the values and goals they pursue as a parent.

Previous intervention studies already showed that value-affirmation methods, which aim to connect persons with their most important and deeply endorsed values and preferences, help to cope effectively with stressful conditions (Cohen & Sherman, 2014). Whereas most of these studies have been conducted with students and in the academic domain, future experimental and intervention-based research could examine

whether this method works with parents and helps them to cope better with stress. If so, value-affirmation methods could too be integrated in the treatment of parental burnout. Hill, Spiegel, and Gennetian (2020) discuss how an adapted, pride-based written self-affirmation exercise increased parents' positive self-concept and their interest in parenting programs and resources, particularly among parents with a high baseline fear of judgment associated with seeking help. Similarly, Sweeney and Moyer (2015) highlight that self-affirmation influences health messages' effect on intentions and behavior. From a preventive standpoint, it could be valuable to include these (repeated) moments of reflection, already in prenatal counseling, in order to pre-arm parents for the challenges that await them in parenthood (Pinquart & Teubert, 2010).

The results for the interaction effects were not completely in line with what we expected, as it was hypothesized that synergistic effects would emerge, implying that the presence of both resources would yield a surplus effect, resulting in even lower levels of parental burnout compared to when only of those two resources would be available. However, for prevention or intervention research, these results are hopeful, as they suggest that, even when parents struggle to form a clear parental identity, targeting their psychological needs (or vice versa) might be enough to prevent them from becoming exhausted as a parents. Of course, similar to well-being not being equal to the absence of psychopathology (Topp et al., 2015), it is important to note that the absence of parental burnout does not imply that parents feel vital and energized in their parental role. Possibly, when looking at more adaptive outcomes, the presence of both a clear parental identity and high psychological need satisfaction are necessary to reach the highest levels of fulfillment, connection and happiness in the parental role.

Limitations and Future Directions

As one of the first studies to simultaneously examine the role of parental identity and parents' need-based experiences, the current study had a number of important limitations that could be addressed in future research. First, the data were collected through an open invitation, which limits the representativeness of this sample. Voluntary participation in an online data collection can lead to self-selection biases. It is possible that only parents interested in parenting issues responded to the survey and that parents suffering from parental burnout in particular were too exhausted to participate. Therefore, it is unclear whether the prevalence of parents facing significant levels of burnout in the current study is representative of the population. In addition, parents from intact families and with higher education levels were overrepresented in our sample, limiting the generalizability of the results. Relatedly, we did not have information regarding parents' ethnicity, so we do not know if there was any racial diversity in our sample. To increase generalizability, future research would do well to recruit parents more actively and to include parents whose ethnicity, family, socioeconomic and educational

conditions are more diverse, for example through parent support services or by doing home visits. Moreover, it would be interesting to select a clinical subsample of parents being at risk for or suffering from parental burnout, examining if the studied psychological resources are even (or even more) important for this group of parents.

The second limitation lies in the use of self-reported measures only. Parents' self-reports may be biased by social desirability. Parental burnout in particular is a characteristic not lightly disclosed by parents in fear of being judged as an incompetent parent. Because parents are expected to raise and encourage their children with unconditional love and affection, some parents may be reluctant to report negative parenting experiences, resulting in an underestimation of the prevalence of parental burnout. The use of self-report measures only in this study may also have led to problems with shared method variance such as an inflation of the strength of associations between the study variables. Future studies should collect data from different sources, for example, through partner reports and via physiological markers such as hair cortisol (Brianda et al., 2020).

A third limitation is the cross-sectional design, which prevents us from drawing conclusions about causality or directions of effects. Most likely, associations between the study variables are bidirectional in nature with perceptions of parental burnout for instance also increasing parental doubts and leading parents to lose direction in their parental role. Furthermore, recently, Arden et al. (2022) found that satisfaction of mothers' psychological needs helps them build stronger inner resources for better or greater self-organization and integrated parental identity development, whereas the frustration of mothers' psychological needs has an energy-depleting effect on their parenting. Therefore, using a longitudinal design would better clarify the direction of effects between parental identity dimensions and their psychological needs on parental burnout.

Finally, as research on parental identity continues to develop, a more in-depth examination of parental identity processes is warranted in future research. For example, whereas this study found that exploration was only to a limited extent a significant moderator of parents' psychological need satisfaction on parental burnout, exploration additionally correlated positively with commitment and ruminative exploration, both of which were significant moderators of psychological need satisfaction on parental burnout. Therefore, future research could gain more insight into the reciprocal associations among these constructs and its meaning in explaining the association between psychological need satisfaction and parental burnout.

Conclusion

Because parental burnout results from a chronic imbalance between demands over resources, it has been argued that prevention and intervention should focus on supporting parents to enable them to restore that balance in their parental role. Whereas past research has focused mainly on identifying risk factors, prevention and

intervention programs would do well to also pay attention to strengthening protective factors. This study contributed to the parental burnout literature and shed light on the interplay between parental identity and psychological need satisfaction, as two psychological resources, uniquely and interactively protecting parents against parental burnout. When future longitudinal and multi-informant research confirms the resilient role of these resources, they could become valuable targets for prevention and treatment of at-risk or burned-out parents. Healthcare professionals working with exhausted parents would then do well to enrich their support with methods aimed at strengthening parents' identity and increasing their energy level through psychological need satisfaction. By doing so, the enriching and joyful aspects of parenthood may again begin to outweigh the (inevitably) exhausting and demanding experiences that come with the parental role.

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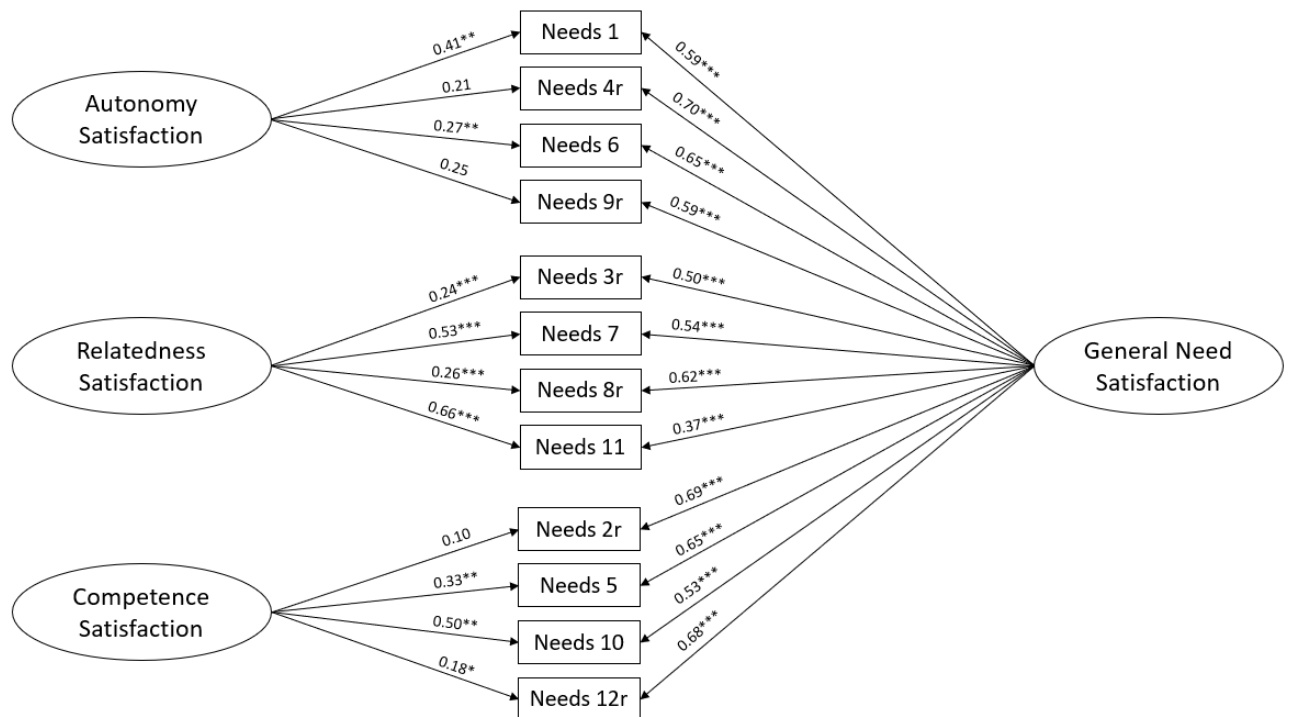
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Figures

Figure 1

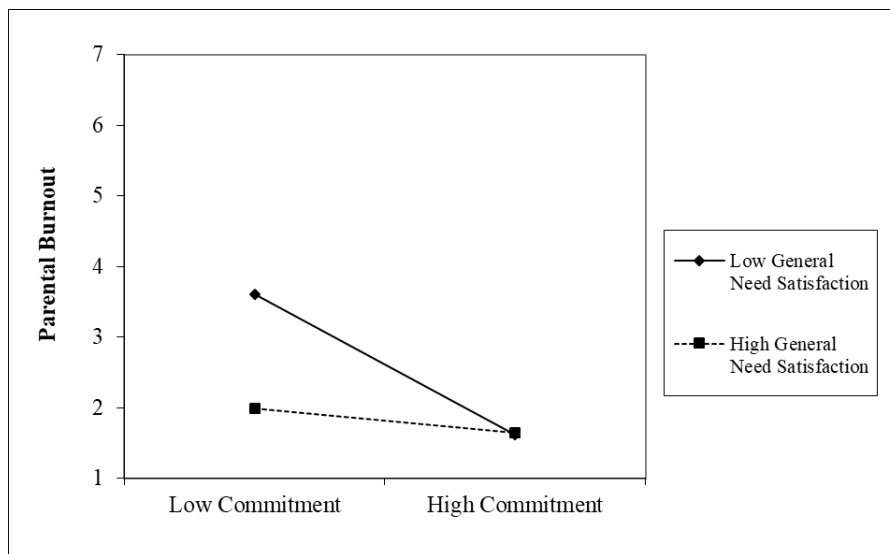
The Bifactor CFA Model of Psychological Need Satisfaction and General Need Satisfaction with Standardized Factor Loadings.



Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. This bifactor model yielded an adequate fit: $X^2(40) = 130.93$; CFI = 0.96, RMSEA = 0.07; SRMR = 0.03.

Figure 2

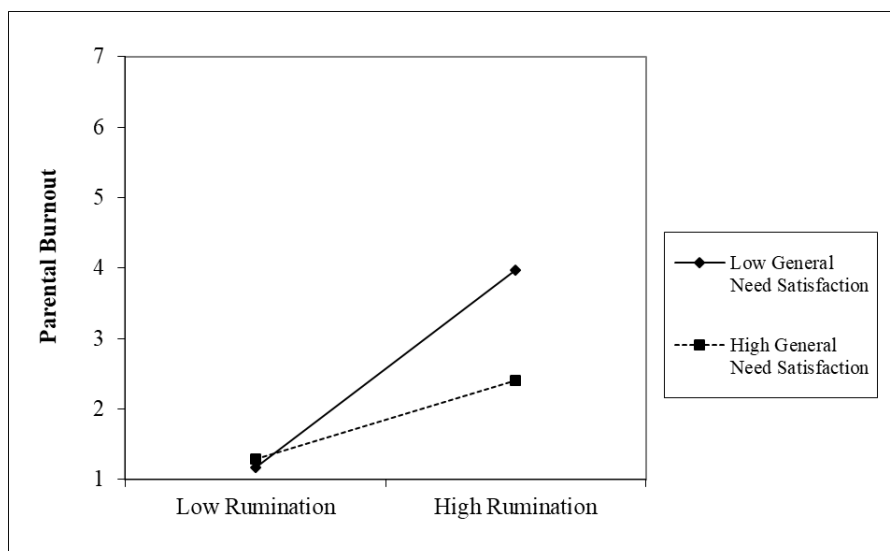
Interaction between Commitment and General Need Satisfaction on Parental Burnout



Note. The negative association between commitment and parental burnout is buffered when parents' general need satisfaction is high. Parents are at elevated risk for parental burnout when they simultaneously display low levels of commitment and experience low levels of general need satisfaction

Figure 3

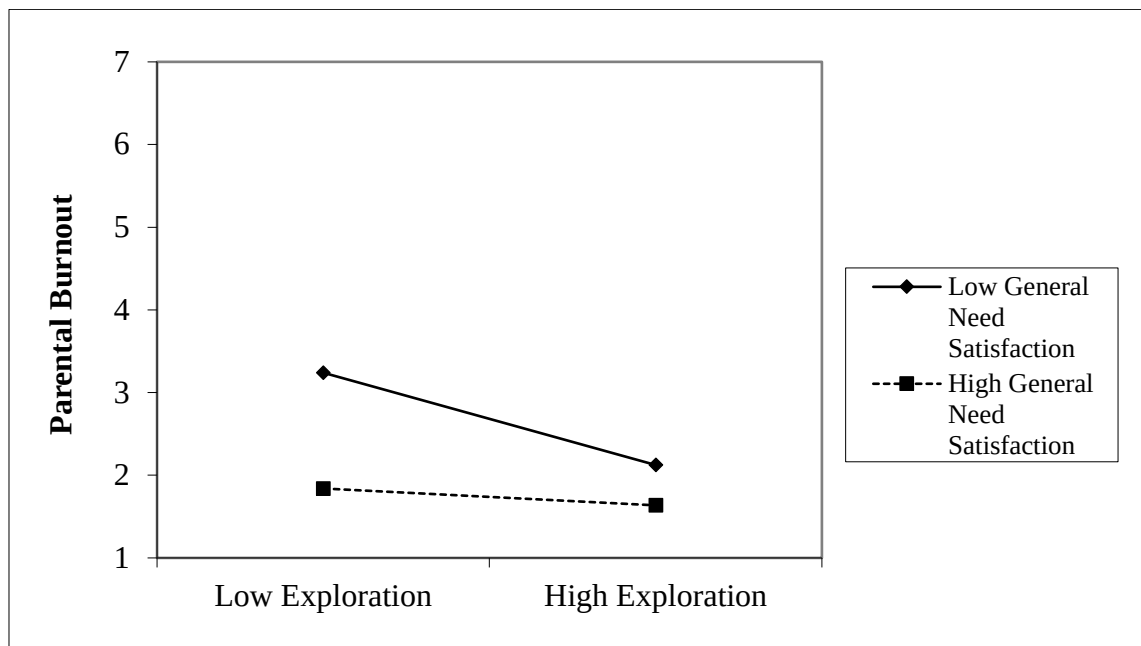
Interaction between Ruminative exploration and General Need Satisfaction on Parental Burnout



Note. The positive association between ruminative exploration and parental burnout is buffered when parents' general need satisfaction is high. Parents are at elevated risk for parental burnout when they simultaneously display high levels of ruminative exploration and experience low levels of general need satisfaction

Figure 4

Interaction between Exploration and General Need Satisfaction on Parental Burnout



Note. The negative association between exploration and parental burnout (although not significant) is buffered when parents' general need satisfaction is high. Parents are at elevated risk for parental burnout when they simultaneously display little exploration and experience low levels of general need satisfaction

Tables

Table 1*Descriptives of and Pearson Correlations between the Study Variables*

	1	2	3	4	5	6	7	8	9	10
1. Gender	-									
2. Education	-0.08	-								
3. Commitment	-0.03	0.10*	-							
4. Exploration	-0.27***	0.11*	0.62***	-						
5. Ruminative exploration	-0.18***	0.16***	-0.09*	0.40***	-					
6. Total Need Satisfaction	0.22***	0.07	0.24***	-0.10*	-0.48***	-				
7. Autonomy Satisfaction	0.17***	0.10*	0.24***	-0.09	-0.41***	0.88***	-			
8. Relatedness Satisfaction	0.15**	0.01	0.20***	-0.01	-0.36***	0.81***	0.55***	-		
9. Competence Satisfaction	0.24***	0.05	0.16***	-0.14**	-0.47***	0.87***	0.67***	0.55***	-	
10. Parental Burnout	-0.23***	0.05	-0.26***	0.11*	0.54***	-0.59***	-0.58***	-0.48***	-0.45***	-
<i>M</i>	-	15.92	3.54	3.47	2.53	3.84	3.58	4.20	3.74	2.21
<i>SD</i>	-	2.65	0.71	0.71	0.75	0.54	0.67	0.60	0.64	1.19
Possible range	-	0-27	1-5	1-5	1-5	1-5	1-5	1-5	1-5	1-7

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. $N = 450$. Gender: 0 = female; 1 = male. Education = continuous variable, the total number of years of education completed by the respondent.

PSYCHOLOGICAL RESOURCES AGAINST PARENTAL BURNOUT

Table 2

Tested Interaction Effects between Parental Identity and General and Specific Need Satisfaction on Parental Burnout. Controlled for Parents' Gender and Educational Level

	β		β		β
<i>Main effects</i>		<i>Main effects</i>		<i>Main effects</i>	
PI-Commitment (COM)	-0.29***	PI-Ruminative exploration (RUM)	0.37***	PI-Exploration (EXP)	0.30
General Need Satisfaction	-0.40***	General Need Satisfaction	-0.27***	General Need Satisfaction	-0.39***
Autonomy Satisfaction	-0.49***	Autonomy Satisfaction	-0.47***	Autonomy Satisfaction	-0.76***
Relatedness Satisfaction	-0.18***	Relatedness Satisfaction	-0.14***	Relatedness Satisfaction	-0.42***
Competence Satisfaction	-0.20***	Competence Satisfaction	-0.01	Competence Satisfaction	-0.27*
<i>Interaction effects</i>		<i>Interaction effects</i>		<i>Interaction effects</i>	
COM*General Need Satisfaction	0.25***	RUM*General Need Satisfaction	-0.16***	EXP*General Need Satisfaction	0.13**
COM*Autonomy Satisfaction	0.35***	RUM*Autonomy Satisfaction	-0.43***	EXP*Autonomy Satisfaction	0.004
COM*Relatedness Satisfaction	0.16***	RUM*Relatedness Satisfaction	-0.09**	EXP*Relatedness Satisfaction	-0.06
COM*Competence Satisfaction	0.19***	RUM*Competence Satisfaction	-0.02	EXP*Competence Satisfaction	0.04

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. $N = 450$. PI = Parental identity

Table 3

Gradients of Simple Slopes of Commitment, Ruminative exploration and Exploration on Parental Burnout for Different Levels of Need Satisfaction (Only Presented for Significant Interactions).

	Gradient of Simple Slopes		Gradient of Simple Slopes		Gradient of Simple Slopes
With Commitment		With Ruminative exploration		With Exploration	
General Need Satisfaction		General Need Satisfaction		General Need Satisfaction	
Low	-1.30***	Low	1.40***	Low	-0.56 ($p = 0.09$)
Moderate	-0.70***	Moderate	0.98***	Moderate	-0.33 ($p = 0.33$)
High	-0.10**	High	0.56***	High	-0.10 ($p = 0.78$)
Autonomy Satisfaction		Autonomy Satisfaction			
Low	-1.53***	Low	2.10***		
Moderate	-0.70***	Moderate	0.98***		
High	0.13***	High	-0.14 ($p = 0.10$)		
Relatedness Satisfaction		Relatedness Satisfaction			
Low	-1.09***	Low	1.20***		
Moderate	-0.70***	Moderate	0.98***		
High	-0.30***	High	0.75***		
Competence Satisfaction					
Low	-1.15***				
Moderate	-0.70***				
High	-0.24***				

Note. Low = moderator value is -1, Moderate = moderator value is 0, High = moderator value is 1.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. $N = 450$.

To ease interpretation, for instance, the interaction between commitment and general satisfaction can be understood as follows: identity commitment predicted lower parental burnout only when general need satisfaction was low (or moderate), not when general need satisfaction was high.

Supplementary File

Supplementary Table 1

Summary for All Constructs in the Measurement Model with All Variables

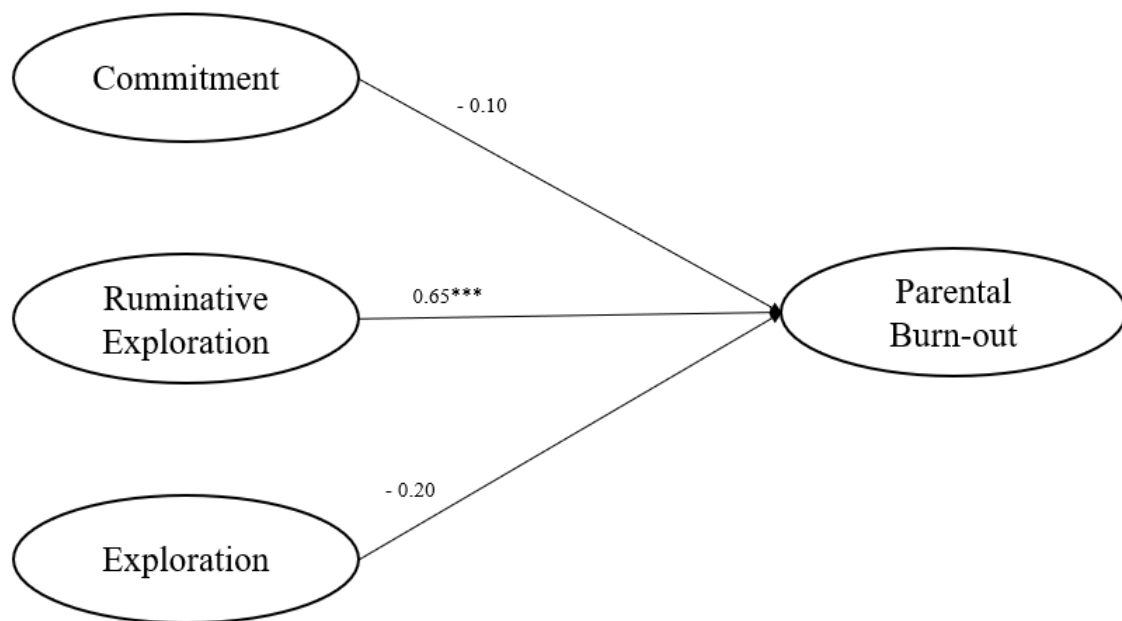
Variable Construct	Items	Factor Loadings	Cronbach Alpha	Construct Reliability	Average Variance Extracted
Parental Identity Commitment	ID 1	0.70***	0.82	0.83	0.54
	ID 7	0.73***			
	ID 10	0.79***			
	ID 13	0.72***			
Parental Identity Exploration	ID 2	0.75***	0.81	0.84	0.56
	ID 5	0.77***			
	ID 11	0.79***			
	ID 14	0.69***			
Parental Identity Ruminative Exploration	ID 3	0.50***	0.79	0.78	0.48
	ID 6	0.82***			
	ID 12	0.65***			
	ID 15	0.76***			
General Need Satisfaction	NEEDS 1	0.62***	0.87	0.48	0.46
	NEEDS 2r	0.65***			
	NEEDS 3r	0.47***			
	NEEDS 4r	0.69***			
	NEEDS 5	0.65***			
	NEEDS 6	0.69***			
	NEEDS 7	0.55***			
	NEEDS 8r	0.63***			
	NEEDS 9r	0.60***			
	NEEDS 10	0.54***			
	NEEDS 11	0.38***			
	NEEDS 12r	0.66***			
Autonomy Need Satisfaction	NEEDS 1	0.22*	0.77	0.19	0.48
	NEEDS 4r	0.30**			
	NEEDS 6	0.10			
	NEEDS 9r	0.32***			
Relatedness Need Satisfaction	NEEDS 3r	0.42***	0.75	0.33	0.43
	NEEDS 7	0.38***			
	NEEDS 8r	0.38***			
	NEEDS 11	0.44***			
Competence Need Satisfaction	NEEDS 2r	0.27***	0.76	0.25	0.48
	NEEDS 5	0.29***			
	NEEDS 10	0.31**			
	NEEDS 12r	0.32**			
Parental Burnout ¹	PBE (9 items)	0.86***	0.95	.95 (total)	0.78
	PBD (3 items)	0.82***	0.83		
	PBF (5 items)	0.90***	0.89		
	PBC (6 items)	0.95***	0.93		

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

¹For Parental Burnout, the four subscales PBE (Emotional Exhaustion), PBD (Distancing), PBF (Feeling Being Fed Up) and PBC (Contrast with Previous Self) were used for the construct.

Supplementary Figure 1

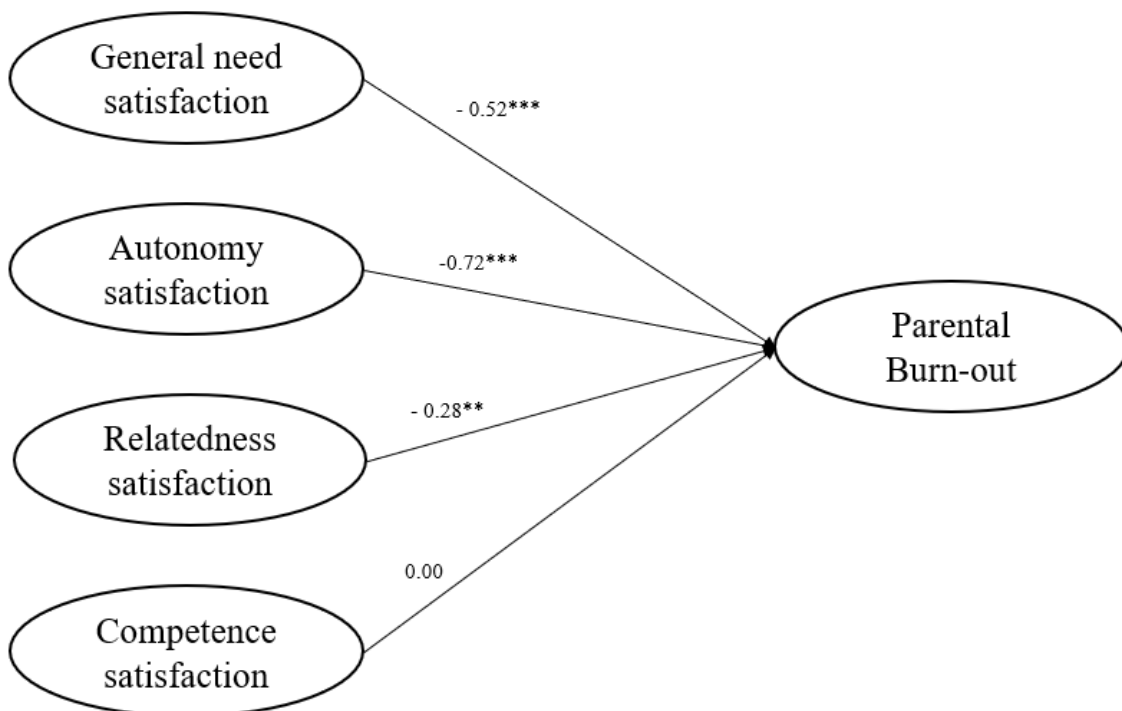
SEM-model with standardized coefficients of the main effects between Parental Identity and Parental Burn-out.



Note. *** $p < 0.001$

Supplementary Figure 2

SEM-model with standardized coefficients of the main effects between Psychological Need Satisfaction and Parental Burn-out.



Note. *** $p < 0.001$, ** $p < 0.01$